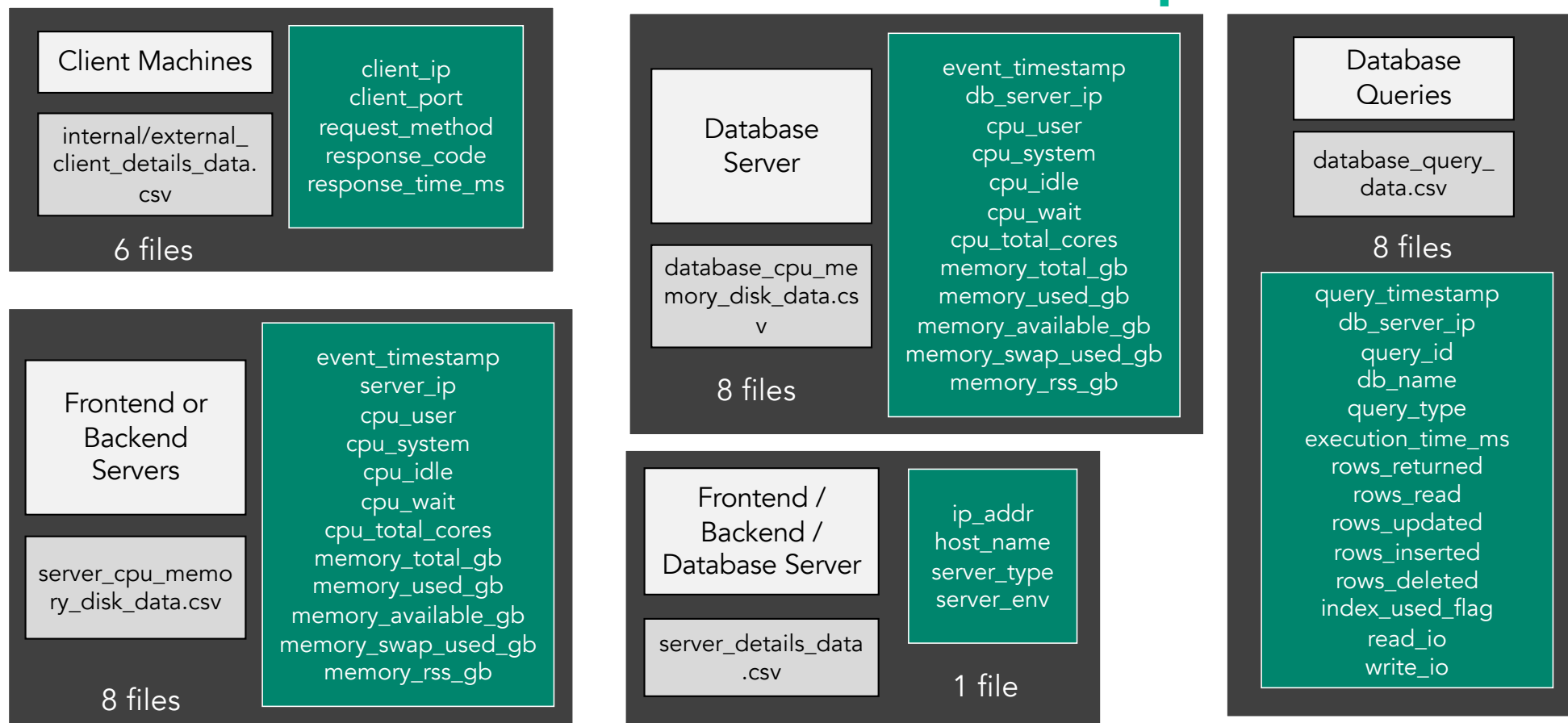


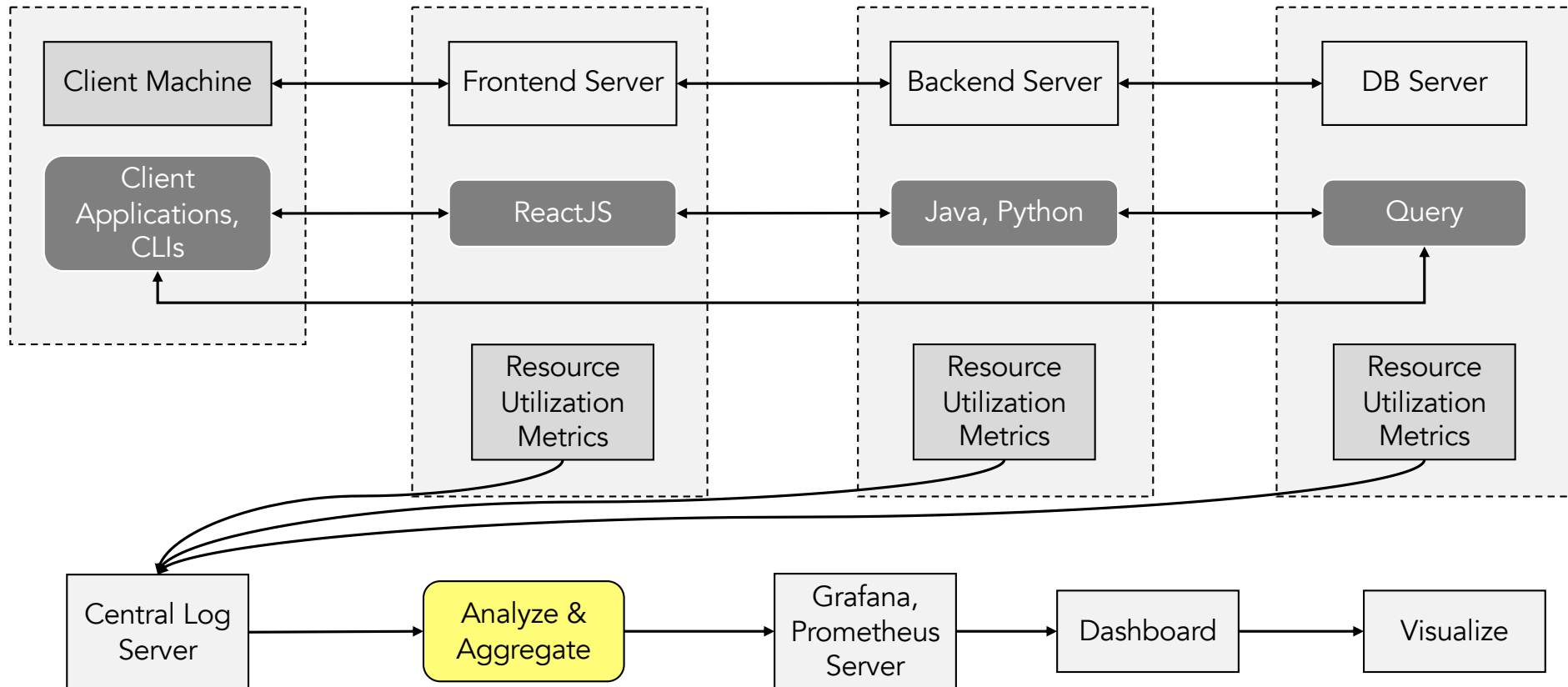
Project 13 Server Log Analysis

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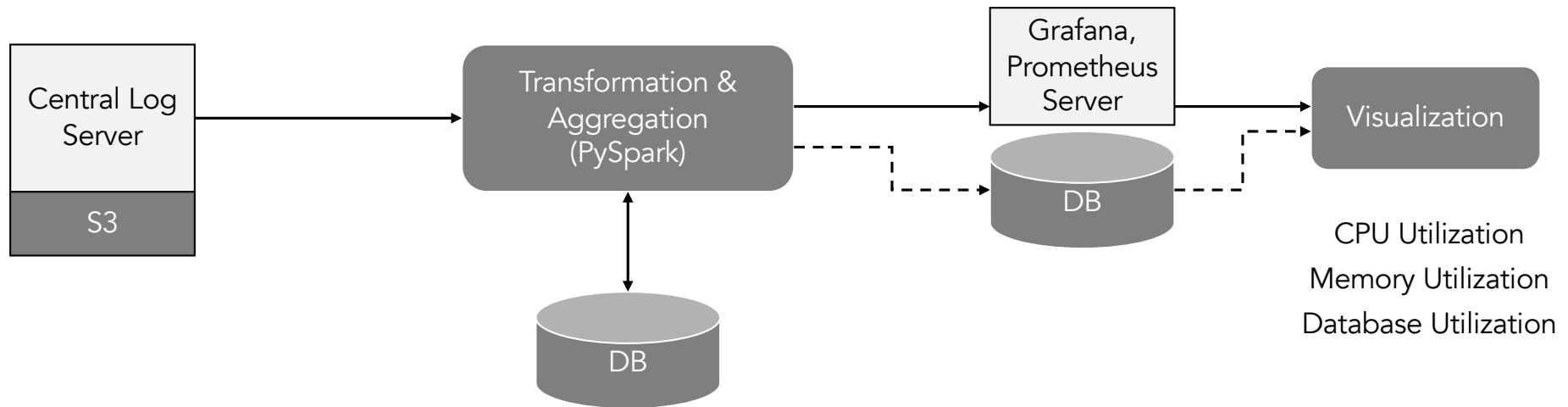
Datasets – 54 GB (Uncompressed)



Proactive & Historical Monitoring



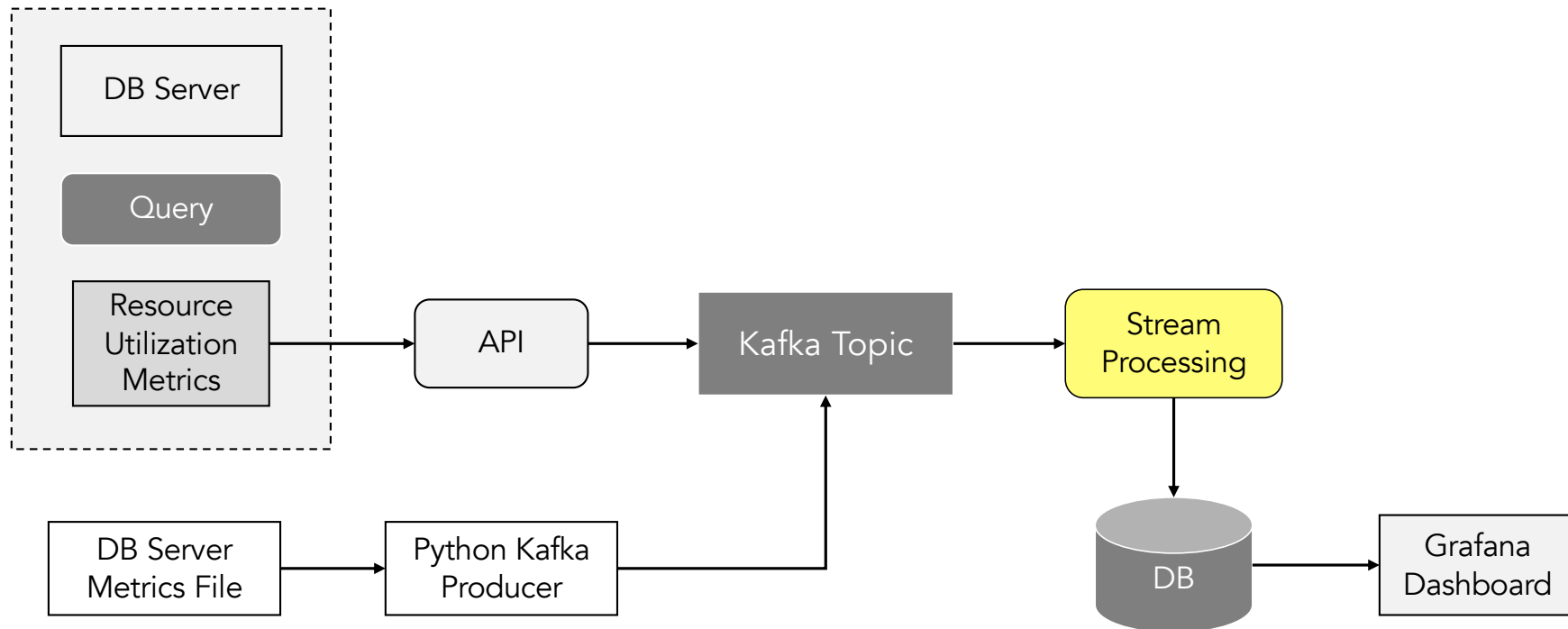
Architecture



Problem Statements – Proactive

- Grafana Reports – 5 mins, 1 hour, 24 hours, 7 days, 1 month, 3 months, 6 months.
 - CPU, memory, disk utilization, database query details etc.
- 1. Inventory of distribution of servers – Total database, application & frontend servers. Does the PROD distribution matches with STAGING servers?
- 2. For the years 2025 and 2026, aggregate data as below:
 - i. Avg. and max. CPU, memory utilization for all servers over a period of 5 mins to 6 months as mentioned above.
 - ii. CPU and Memory spikes for Prod servers – over 1 week and 1 month.
- 3. Database Server analysis
 - i. Avg. query execution time for each database.
 - ii. What is the avg. and max. execution time of queries.
 - iii. Analyze why some SELECT, UPDATE & DELETE queries are taking more time than the others. What is their CPU, memory, read & write IO as opposed to other queries. Is there any correlation with INDEX usage.
 - iv. Find out the execution time and resource utilization of queries that use index against the queries that don't use index.
 - v. In which scenarios swap memory is used the most.
 - vi. What is the ratio of rows read vs rows returned of queries that are using index as opposed to queries that don't.

Reactive Monitoring – DB Server



Problem Statements – Reactive

- Query & Resource Monitoring for all query types (SELECT, UPDATE, DELETE, INSERT).
 - Read records from the files - database_query_data_5.csv & database_cpu_memory_disk_data_5.csv
1. Monitoring should be for every 15 mins window.
 2. The SLA for SELECT queries is 2 seconds. Capture the queries whose execution time is more than 2s and their corresponding data like CPU, memory, read io, rows read, rows returned etc. Which CPU usage is high (user, system, wait) for these queries? Has there been any swap usage by these queries? What is the resident memory (rss) usage?
 3. SLA for UPDATE & DELETE queries is 2 seconds. Capture the queries with more than 2s execution. What is the no. of rows updated and read. What is read and write IO for these queries?
 4. What is the avg. and max. execution time of INSERT queries along with their IO and CPU usage.
 5. What is the avg. CPU idle time for Select, Update & Delete queries when indexes are getting used.
- Output should be saved in a MySQL database with all the above information, so that it can be shown in Grafana dashboard.